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Mapping quantum industry demands to education: a critical analysis of skills, qualifications, and modalities

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Abstract

Quantum technologies and computing are an emerging area which offers a new paradigm to solve complex problems using the principles of quantum mechanics, where classical computing faces limits. Due to the advantages of quantum computers, today, there are several industries focusing on different aspects of quantum technologies based on their physics to explore the most efficient and useful platform for implementing applications. Since the scope of the quantum companies is diverse, it is important to understand the education, skills, and qualifications required for different job roles, as this will aid global educational institutions in constructing concentrated disciplines in this field. This paper provides a detailed critical analysis of different job descriptions for education, skills and qualifications. Most of the qubit modalities, such as superconducting, semiconducting, topological, nitrogen-vacancy centres, ion-traps, neutral atoms, and photonics, have been covered. Additionally, quantum software domains such as quantum machine learning, cryptography and error corrections have been discussed with fields such as quantum sensors and metrology. Finally, based on the patterns, recommendations are given to enable better preparation of skills and infrastructure for educational institutes and individuals who would like to pursue a career in the field of quantum technologies.

Keywords: Quantum technologies; Quantum education; Quantum industry analysis; Quantum job market analysis; Quantum career pathways

1 Introduction

There is a misconception regarding the terms quantum computing and quantum mechanics. Rather than terming the history of quantum computing, we must pronounce it as the history of quantum mechanics because the concept of quantum computing evolved with the idea of Richard Feynman in the 1980s in his famous speech titled Simulating Physics with Computers [1]. The development of this field started in the early 20th century with more intensity. With the contributions of physicists like Planck, Einstein, Bohr, De-Broglie, Werner Heisenberg, Max Born and Dirac, quantum mechanics as a field emerged, and over the course of the years, around the 1980s, Feynman had a vision to simulate physics with the help of computers. This sparked the idea of quantum computing as a field [1].

Since then, contributions to famous quantum algorithms like Deutsch [2], Grover's search [3], and Shor's algorithm [4] have been developed. There were major contributions happening in quantum hardware, quantum information processing between 2000-2019 and that too mostly in university labs. But the turning point occurred between 2017-2019 [5], when the quantum wave began, and many startups started booming. Gradually, universities started adopting the course, and the government were investing funds in the research and development of the quantum field. This is our observation on the development of the quantum field.

Since then, there has been substantial funding from government and private players, startups, and other big organisations that have started hiring PhDs [6, 7]. To improve the situation to a certain extent, teachers and professors have started exploring AI tools to introduce physics and its concepts [8–10]. The main concern here is that quantum is still in the novice stage despite successful breakthroughs every now and then. However, there is less awareness about different quantum domains, their careers, and the skill sets required in particular quantum domains. We understood this issue by giving mentorship and career guidance to students and early professionals. This led to the main motivation of writing the paper, along with 2 other reasons. First, during multiple student outreach activities—most notably the qiskit fall fest and various career guidance sessions—the authors interacted with large numbers of undergraduate students interested in quantum computing. Many of these undergraduate STEM students expressed unawareness about the skill sets required to pursue specific domains within quantum technologies. They frequently raised concerns about the lack of clear guidance on how to align their learning pathways with industry expectations. Second, a review of existing literature [11, 12] further reinforced these observations. The author stressed the importance of building an understanding of quantum and quantum careers as early as possible, and students in their survey raised concerns regarding the uncertainty of pursuing quantum education because of the lack of skill awareness one must develop to fit into the quantum industry standards. Undergraduates are also encouraged to develop critical thinking, as highlighted here in [13, 14].

In this study, we examine the quantum job market by analysing the job descriptions across various sub-fields of quantum science discussed in this work. Our primary objective is to identify the key skills required for students to align their profiles with industry expectations. Additionally, we have explored the academic qualifications and degrees commonly sought out by employers, thereby providing guidance for students aiming to pursue careers in specific areas of the quantum field.

2 Theory

Before we discuss the results of this paper, we highlight some of the current industrial domains in quantum technologies that are being pursued by organisations globally and also have a track record of persistent research in academic labs. These domains are considered in this paper for analysing the job descriptions they post and the skills that are required of the candidates.

2.1 Quantum hardware platforms

2.1.1 Superconducting qubits

Superconductivity was first discovered in 1911 by Kamerlingh Onnes [15], which became a major breakthrough today. Whenever current flows, there is always some resistance present. But when a certain material is cooled below its critical temperature, there is no resistance when current flows, and this process is called superconductivity. Mercury was the first element to display superconductivity at 4.2 K. From then series of experiments were conducted that proved the existence of superconductivity at lower temperatures. As there is low-temperature superconductivity, there is also high-temperature superconductivity. This doesn't mean superconductivity occurs at room temperatures, but significantly higher than in the low-temperature superconductors. For instance, the critical temperature (T_c) for mercury (low temperature superconductor) is 4.15 K, but Lanthanum Barium Copper Oxide [16], which is the first high temperature superconductor with T_c to be 30 K.

2.1.2 Semiconducting qubits

Semiconductor spin qubits are a type of qubit that leverages the intrinsic properties of electrons confined in semiconductor materials like silicon [17] or gallium arsenide (GaAs) [18]. Electrons have an intrinsic property of spin up and spin down, and the quantum information is encoded in the spin of a qubit (either 0 or 1). There are four different types of spin qubits, such as single-spin qubit, donor spin qubit [19], singlet-triplet spin qubit [20], and exchange-only spin qubit [21], each utilising different methods for encoding and manipulating spin states.

In semiconductor spin qubits, the electrons are confined in quantum dots. Quantum dots are tiny semiconductor structure that traps electrons in all three spatial directions. The spin of the trapped electrons is controlled by applying electric fields via gate electrodes. The first spin qubit quantum computer was proposed by Daniel Loss and David P. DiVincenzo in 1997 [22]. Few conditions in DiVincenzo's criteria need to be satisfied for a quantum computer to be considered scalable, and spin qubits are said to meet most of the criteria [23]. One of the major advantages of spin qubits is scalability because fabrication techniques are well-established in the industry, which has the potential to scale up to a large number of qubits. However, a significant disadvantage is the difficulty in directly measuring the spin state of an electron.

2.1.3 Nitrogen vacancy center

Nitrogen vacancy (NV) centres in diamond [24] qubits are atomic-scale defects where a nitrogen atom sits adjacent to a vacancy in the diamond lattice, yielding an electron spin that can be optically initialised, manipulated, and read out at room temperature. Their robust quantum coherence, even under ambient conditions, makes them ideal for both quantum computing [25] and high-precision quantum sensing applications [26].

2.1.4 Ion-traps

Trapped ions have always been a lucrative candidate for quantum technologies, especially in computing, because of the individual atomic control and precision [27]. Apart from computing, they find applications in high-accuracy mass spectroscopy [28], atomic clocks [29], and fundamental physics-based experiments [30]. Ion traps confine ions using carefully engineered combinations of static and oscillating electric and/or magnetic fields, creating a localised potential well that spatially restricts ion motion. There are two types of traps - Paul [31] and Penning [32], where the former RF fields dynamically stabilise ions and the latter relies on a static magnetic field combined with an electric field for trapping. Ion cooling techniques reduce thermal motion, enabling extended interrogation times and ultra-high measurement resolution, making ion traps essential tools in modern precision science.

2.1.5 Neutral atoms
Neutral atom quantum computing systems make use of laser cooling [33], magneto-optical trapping [34] and optical tweezers [35] to trap and hold atoms at a particular location in space. These systems exploit Rydberg states of neutral atoms to create quantum entanglement protocols and high-fidelity multi-qubit gates. The atoms offer scalability and allow the preparation of dense, defect-free atom configurations over two- or three-dimensional geometries. Applications include quantum simulation of many-body systems [36], phase transitions in matter [37], quantum algorithms for machine learning [38], error correction [39] and quantum sensors [40].

2.1.6 Photonic qubits
Photonics is the discipline of how electrons and photons work together to create new applications and devices. In general, quantum photonics is the study of how light particles, such as photons, behave and interact at the quantum level. Photonics-based quantum computers work by generating, guiding and manipulating photons on tiny optical chips by utilising waveguides and beam splitters [41–43]. This enables photons to represent quantum bits (qubits) via features such as polarisation or phase, which can exist in several states at once. Controlling how these qubits interact allows the machine to do complicated calculations considerably faster and more efficiently than traditional systems [44, 45]. Beyond computing, photonics has a wide range of possible applications, including ultra-secure quantum communications, improved imaging systems and precise medical or environmental sensors.

2.2 Quantum machine learning
In classical machine learning, a parametrised model learns patterns from training data with known (true) labels given to it using statistical techniques. The process usually involves providing the training data samples to the model, the model performs certain mathematical operations, and then gives an output prediction that is used in the cost function. The cost function takes the predicted values and the true values to compare the error, i.e. deviation of the predictions from the true values. This error is then given to an optimiser in order to adjust the parameters of the model to increase the accuracy of predictions of the model.

However, in quantum machine learning (QML) [46–51], the input data is first converted into quantum states using various encoding mechanisms such as amplitude, phase or basis scheme, and parametrised quantum circuits (PQC's) act as quantum models that process this quantum input data. Quantum measurements of these quantum circuits result in a binary outcome. Using these binary numbers, expectation values are calculated, and this forms the cost function. Furthermore, the optimiser takes the output of the cost function and the current parameters of the PQC in order to calculate new parameters for the PQC that are updated accordingly.

Two important machine learning algorithms, namely Neural Networks [52] and Support Vector Machines [53], have heavily inspired the field of quantum machine learning because of the similarity of those algorithms with the variational quantum circuit models. The architecture of a typical machine learning model is used directly in QML. QML differs from the classical way in the sense that encoding schemes are utilised to convert classical inputs into quantum states, and the classical ML models are replaced by PQC's.

2.3 Quantum finance

In macroeconomic terms, finance refers to the flow of funds in an economy. This includes various aspects such as allocation of resources, distribution of income and management of risk. An increase in the economic size and population of a country significantly increases the complexity of money management. Some of the major challenges faced by the finance sector are uncertainty and risk, portfolio optimisation, fraud detection, derivative pricing, market prediction and regulatory compliance.

Quantum mechanics has shown some useful impact in the finance sector. One such example is the application of lattice gauge theory to a foreign exchange problem [54]. Coming to the main challenges, risk management is the process of recognising possible financial hazards, assessing their implications, and putting plans in place to successfully reduce or control those risks. For example, in [55], one of the prominent quantum algorithms mentioned for risk management is the quantum Monte Carlo algorithm. This algorithm helps identify patterns and relationships in complex financial data, offering better predictions for risk scenarios. In derivative pricing, quantum Monte Carlo simulations provide exponential speed-ups in sampling complex probability distributions, making derivative pricing more efficient and cost-effective. One such example is Baaquie's Black–Scholes–Schrödinger equation [56] that bridges quantum physics and finance, offering a novel approach to pricing derivatives.

2.4 Quantum cryptography

The main goal of cryptography is to transmit data securely. Quantum cryptography is a rapidly evolving field that uses the principles of quantum mechanics to develop cryptographic protocols resistant to both classical and quantum computational attacks. The origin of cryptography can be traced back to the 1980s, where Stephen Wiesner published *quantum conjugate coding* [57]. It acted as a foundation for the first Quantum key distribution (QKD) system, which is the heart of quantum cryptography called the BB84 protocol, which is designed to have secure communication [58]. The main reason behind the establishment of quantum cryptography as a field is due to the breakthrough by Peter Shor, who created a quantum algorithm with which modern encryption schemes like Rivest, Shamir, Adleman (RSA) and Advanced Encryption Standard (AES) can be decrypted within minutes. In order to achieve that, companies must have a fully functional quantum computer with fault-tolerant qubits which can run Shor's algorithm. Unfortunately, current technology is not advanced enough to accommodate both.

After the breakthrough of Shor's algorithm in 1994, there has been an increase in research work in the field of quantum cryptography. Beyond QKD, other applications such as quantum digital signatures and quantum secret sharing were developed to authenticate and protect information [59]. Furthermore, quantum authentication protocols have been developed to verify the origin and integrity of data through quantum states [60]. These improvements demonstrate the adaptability of quantum cryptography, tackling multiple facets of secure communication beyond mere key distribution. Despite the potential of quantum cryptographic methods, there are a few persistent challenges in implementation. One of the challenges is the environmental disturbance in which the quantum system is highly sensitive, questioning the stability of qubits. Another challenge is the fully functional quantum computer, which can simulate complex calculations [61].

Furthermore, the expense and intricacy of establishing the requisite quantum infrastructure, including quantum repeaters and detectors, constrain the scalability of these systems. Researchers persist in investigating methods that could not only address the mentioned challenges but also integrate with classical systems to create hybrid models that balance security and feasibility [62], and that's how post-quantum cryptography has been developed, making classical algorithms resistant to quantum attacks. These algorithms do not depend on quantum hardware but are engineered to endure the processing capabilities of quantum computers, fulfilling the urgent requirement for safe cryptographic solutions while extensive quantum networks are still under development [63]. Quantum cryptography and post-quantum cryptography are complementary strategies for safeguarding future communications against the progression of quantum technology, underscoring the necessity for ongoing study and innovation in this field.

2.5 Quantum error correction

The origin of quantum error correction (QEC) [64] is rooted in quantum information theory (QIT), addressing the unique challenges of preserving quantum information. In quantum systems, information is stored in fragile quantum states that are highly susceptible to errors during all stages of information handling, such as storing, processing, manipulating, and measuring. These errors arise due to decoherence, environmental noise, imperfect hardware architecture, and crosstalk between qubits. To mitigate these issues, several quantum error correction codes (QECCs) have been developed. For instance, errors in storing information, such as decoherence, are addressed by encoding a logical qubit into multiple physical qubits, a process known as redundant encoding (e.g., Shor's code encodes 1 logical qubit into 9 physical qubits).

Errors in processing information, like gate errors, can be corrected using fault-tolerant quantum gates and stabiliser codes. Errors in manipulating information, such as bit-flip and phase-flip errors, are handled through codes like Shor's code [4] and Steane's code [65], often aided by syndrome measurements. Measurement errors are addressed by detecting faults in physical qubits rather than directly measuring logical qubits, preserving their quantum state. The significance of QEC lies in ensuring that quantum information cannot be copied or duplicated, unlike classical information, forming the basis of quantum cryptography and secure quantum communication protocols. Quantum Error Correction is a multi-disciplinary field including quantum computing, cryptography, communication, storage, sensing, and networking.

2.6 Quantum sensors

The definition of quantum sensors can be divided into two divisions. One is quantum sensing, and another is a quantum sensor [66]. The main objective of quantum sensing is to detect and measure various physical quantities like magnetic fields, time, and acceleration. Conversely, quantum sensors typically utilise phenomena such as particle spin or atomic and ionic energy states to detect environmental changes. A primary advantage of quantum sensors is their capacity to exceed classical limitations, including the usual quantum limit, which dictates the precision of measurements obtained by classical methods. A quantum sensor can identify minute variations in gravity or weak magnetic fields, facilitating applications in medical imaging, navigation, and materials science. Technologies like atomic clocks, which serve as quantum sensors for time, and magnetometers, which identify magnetic fields, illustrate the enhancement of precision measurement through quantum mechanics.

3 Methodology

Between October 2024 and March 2025, we surveyed a broad sample of different industry job postings from the careers web pages of 37 companies working in the quantum technology sector. These postings were gathered from multiple sources, including company career webpages as well as professional networking and job platforms (e.g., LinkedIn and Naukri). The quantum companies surveyed here are based out of the UK, USA, Canada, Australia and the European Union because of the unavailability of data on quantum companies from other countries, which is the geographical limitation of this paper. We focused on several key quantum domains to ensure coverage of distinct subfields of the industry. Job listings were obtained for roles related to domains like different quantum hardware platforms, quantum machine learning (QML), quantum finance, quantum cryptography, and quantum error correction (QEC). Additionally, we also discussed industry trends in quantum sensing and quantum metrology. Within each domain, we explicitly searched for career opportunities on the company's career page and LinkedIn job postings. Duplicate postings (for example, the same advertisement appearing on multiple platforms) were removed so that each job description in the dataset was unique. We have selected the previously mentioned domains and qubit types based on the ongoing research and their potential for future development. Currently, the type of qubits that are actively being explored in the industry are superconducting, semiconductor, ion trap, neutral atom, photonic, NV centre and topological. Each of those has its own strengths and suitability for different quantum applications. The active development and progress of these domains will positively influence the field of quantum technologies.

Since quantum is in its initial stages of research, some companies did not have job offerings. In that case, we analysed the previous job postings that no longer accept applications, and we used them to analyse the skills, qualifications, and degree required. For QML, quantum finance, nitrogen vacancy centre and topological qubits, we have given a general description of skills due to the limited availability of roles (lack of statistics) and their similarity with other roles. In total, we collected N = 90 unique job descriptions (where N is the total number of postings collected across 37 different companies) for analysis. We employed qualitative research methods [5] to examine the job descriptions. Each job posting was thoroughly read, and any mention of a required skill or competency was noted. For each job description, every unique skill mentioned was counted once (even if the same skill appeared multiple times within a single posting, it was only recorded a single time for that posting to avoid over-counting). This process yielded a comprehensive list of distinct skills across all collected job postings. Also, we noted down the required degree and qualifications. Once all job descriptions were analysed based on skills, degree and qualifications, we used a summative content analysis approach to count the number of skill occurrences in job descriptions to identify their frequency and quantify their relative importance. From these counts, we then calculated the percentage share of each skill/degree/qualification relative to the total number of mentions observed across all postings. The formula used for calculation is,

$$\text{Skill / Degree / Qualification Percentage} = \frac{\text{Frequency of a given skill/degree/qualifications}}{\text{Total frequency of all skills/degree/qualifications}} \times 100\% \qquad (1)$$

From Equation (1), the frequency of skills/degree/qualification means the number of job descriptions in which that skill was mentioned. Total frequency of all skills/ degree/ qualification is the sum of the frequencies of all individual skills/ qualifications /degrees mentioned. Also, we used the term *mentions*, which means the total number of skills/ qualifications/degrees noted after analysing all the job descriptions. Calculating the percentage according to the total mentions helped gauge on-demand skills in the quantum industry job market. All coding and frequency tallies were conducted using a spreadsheet tool (Google Sheets) for accuracy and traceability. We also plotted the visualisation of results as bar graphs and pie charts. Once we get the big picture of the required skill set, qualifications and degree, we will discuss briefly how each skill is interdependent and the need to focus on developing those industry-ready skills during PhD, master's or while working in other non-quantum-based industries. The outcome of this analysis highlights the notable trends across the quantum domains.

4 Results and discussion

Quantum hardware refers to the design, organisation, and implementation of physical components to manufacture a quantum computer. Here, physical components consist of qubits, quantum gates, control and measurement systems, cryogenic systems, interconnects and wiring, optical components, etc. Within these components, qubits play a major role. Depending on the qubit architecture, applications of a quantum computer are varied. For example, a quantum computer manufactured with a superconducting chip is called a Superconducting quantum computer. Fig. 1 [67] shows different types of physical qubit

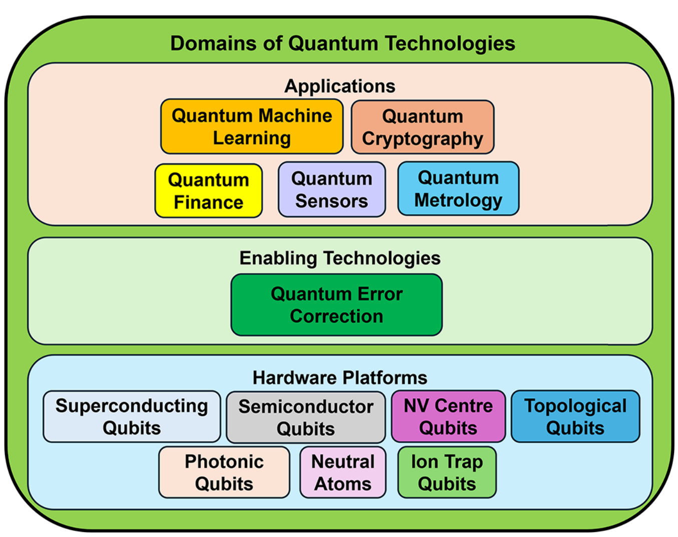

Figure 1 Domains of Quantum Technologies

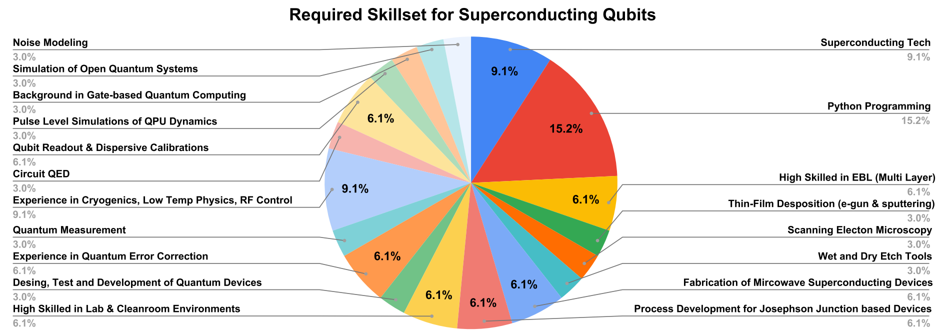

Figure 2 Required skillset in superconducting qubits architecture

realisation. Every physical qubit has a set of advantages and limitations. The scope of this paper is neither to discuss the advantages nor the limitations.

4.1 Superconducting qubits

4.1.1 Required skill set in superconducting qubit

We analysed 6 job descriptions of a total of 33 various mentions of skillset from 4 companies. From Fig. 2, we can understand that superconducting qubits follow a tightly interconnected workflow where each phase builds on another one with superconducting technology knowledge (3 mentions or 9.1%) and circuit QED background (1 mention or 3.0%) being crucial. The process begins with fabrication skills such as thin-film deposition (1 mention or 3.0%), where superconducting materials are precisely layered onto substrates and multi-layer electron-beam lithography or EBL (2 mentions or 6.1%), forming the fundamental structure of the quantum processor. After deposition, advanced patterning techniques like electron beam lithography are employed to pattern circuits at nanoscale precision, particularly for crafting critical structures such as Josephson junctions (2 mentions or 6.1%), wet and dry etching (1 mention or 3.0%) techniques precisely remove unwanted materials, and define accurate geometric patterns throughout the fabrication process. To carry out the fabrication and characterisation of superconducting circuit devices, it is essential to have high skills in lab and clean room environments (2 mentions or 6.1%).

Scanning electron microscopy (1 mention or 3.0%) provides essential high-resolution imaging, allowing verification of the accuracy and integrity of the fabricated circuits, all of which are necessary for producing high-quality Josephson junctions and microwave superconductors. This leads to the creation of processes for Josephson junctions (2 mentions or 6.1%) as well as the manufacturing of microwave superconducting devices (2 mentions or 6.1%), which are then integrated into functional circuits. Next, cryogenics, low-temperature physics and RF control systems (3 mentions or 9.1%) ensure that qubits operate at millikelvin temperatures, which is required to preserve superconductivity and quantum coherence and to drive qubit gates, manage coupling, and handle noise, with noise modelling (1 mention or 3.0%) contributing to improved coherence. Candidates having knowledge on gate-based quantum computing have the advantage of understanding the experiments mentioned previously (1 mention or 3.0%).

Accurate qubit readout and dispersive calibration (2 mentions or 6.1%) are critical for state measurement because they directly affect quantum operations and error correction. Finally, quantum error correction (2 mentions or 6.1%), open quantum system simulations

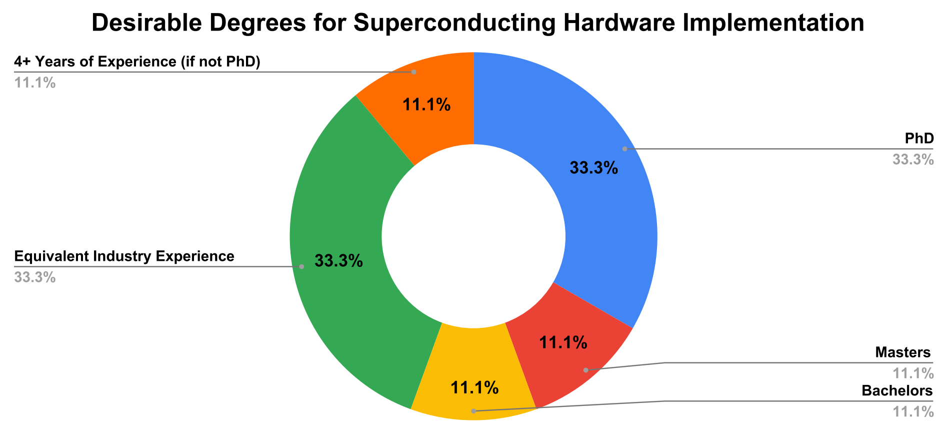

Figure 3 Expected Degree in Superconducting qubit hardware Implementation

(1 mention or 3.0%) and pulse-level simulations of QPU dynamics (1 mention or 3.0%) and quantum measurements (1 mention or 3.0%) integrate all components, ensuring scalability and dependability while minimising errors caused by decoherence and gate defects. The design, test and development of quantum devices (1 mention or 3.0%) plays an important role where programming can help. This interdependent process, which includes Python programming (5 mentions or 15.2%) for operations such as automation, control, and data analysis, is critical for the progress of superconducting quantum technologies.

4.1.2 List of desirable degrees in superconducting hardware architecture

Fig. 3 shows the importance of a PhD degree (3 mentions or 33.3%) as well as equivalent Industry experience (3 mentions or 33.3%), which we can assume as relevant skill sets without a PhD degree, but at least holding a masters degree (1 mention or 11.1%). A bachelor's degree is also mentioned, but the percentage of hiring will be extremely less (1 mention or 11.1%). There was a single mention of 4+ years of experience (1 mention or 11.1%) if the candidate does not have a PhD. These were calculated by considering 9 mentions of the criteria. Importance must be given to completing higher education because any type of experimental realisation of qubits requires special expertise in the fabrication process.

4.1.3 Desirable qualification in superconducting hardware architecture

Superconducting qubit research necessitates a wide range of capabilities, with companies requiring expertise in one or more domains depending on the individual function. Here, the total number of mentions is 13 for desirable qualifications. Physics (5 mentions or 38.5%) is the most in-demand qualification because it provides the theoretical foundation for understanding quantum mechanics, superconductivity, and qubit dynamics, making it necessary for careers in design and research. Electrical engineering (3 mentions or 23.1%) is essential for creating practical systems such as RF control, microwave engineering, and cryogenic electronics, which are required to operate and control qubits. There was a single mention of other engineering fields, which accounts for 7.7%. Material Science (2 mentions or 15.4%) is concerned with the construction and optimisation of superconducting materials, assuring the formation of reliable Josephson junctions and avoiding decoherence, making it vital for roles in device fabrication. Experimental Physics (2 mentions or

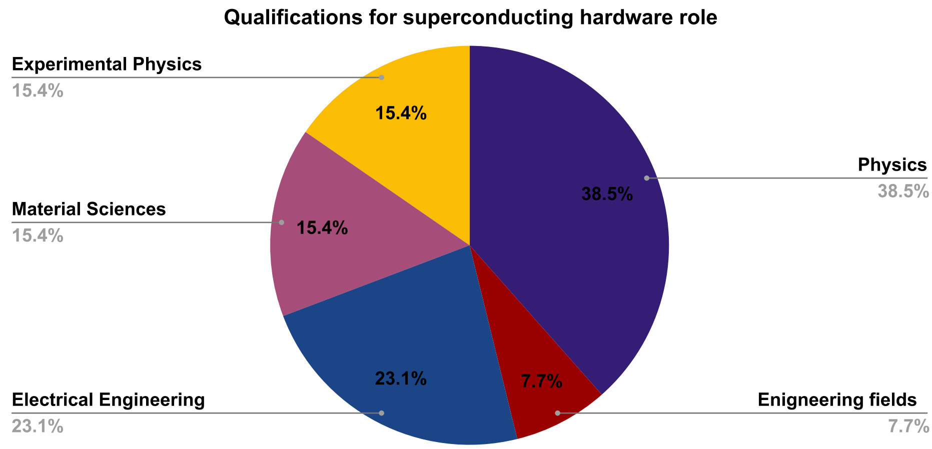

Figure 4 Expected qualification in Superconducting qubit hardware Implementation

15.4%) combines theory and application, emphasising expertise in low-temperature experiments, qubit testing, and calibration. Companies frequently favour applicants who possess one of these qualifications as mentioned in Fig. 4, as each serves a distinct and complementary role in the advancement of superconducting quantum technology.

4.2 Topological qubits

Topological qubits take advantage of quantum mechanical exotic particles called non-abelian anyons to encode information, enabling topological protection of qubits from local noise such as decoherence. Microsoft is the only company pursuing research in topological quantum computing that recently announced their Majorana 1 chip [68–70]. The skills required for getting a job in the field of topological qubits are similar to those of superconducting qubits. However, in addition to those skills, it is helpful to have an understanding of topological physics and some part of abstract algebra to appreciate the rich theory behind topological phenomena.

4.3 Semiconductor spin qubits

4.3.1 Desired skills in spin qubits

There is a range of technical skills that are sought in a candidate who wants to join the quantum spin qubit-based industries. We discuss the important skills that count in this section from 12 job descriptions across 5 different companies, with a total of 39 mentions of different skills. From Fig. 5, it is observed that cryogenics, electrical measurement techniques and vacuum technology are 25.6% of the total skill share (10 mentions). It involves knowledge regarding operation of dilution refrigerators (loading samples, managing cool down sequences), familiarity with cryogenic hazard safety protocols, low-temperature weak signal measurement, ability to design and implement custom circuits or measurement protocols that address issues like impedance mismatch, cross-talk, or electromagnetic interference, operation, and troubleshooting of vacuum pumps (such as turbo molecular, ion, or cryopumps), and integrating vacuum environments with deposition tools or etching systems. With 18.0%, analogue/digital integrated circuit design, ASIC, PCB design, and RF systems form the next essential skill (7 mentions).

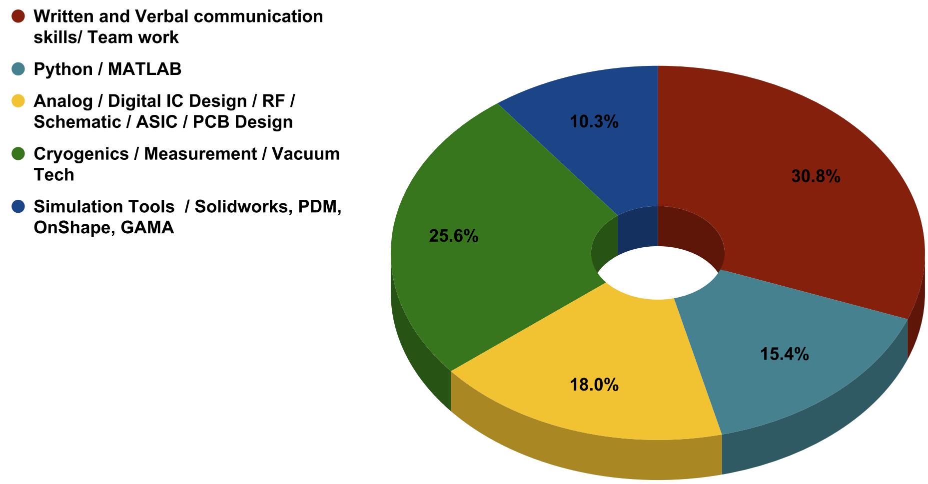

Figure 5 Desired skills in spin qubits

Typically, candidates bring experience in low-noise signal amplification, filtering and signal conditioning, developing digital controllers for signal timing and synchronization, customized optimization of circuits for low power consumption, low noise performance and high-speed signal processing, PCB design experience using Cadence, Altium, Mentor Graphics, Synopsys and RF circuit design, signal processing and reflectometry techniques for measurements. Programming language experience using either Python or MATLAB (6 mentions or 15.4%) is important for data analysis, visualisation, running simulations, modelling, automation and instrument control and software development of quantum systems. Apart from these skills, it is also necessary for candidates to have some electromagnetic simulation experience using tools like COMSOL, Ansys, HFSS or CST microwave studio and custom designing complex mechanical components and assemblies for experimental apparatus using Solidworks, OnShape or CATIA (4 mentions or 10.3%). Tools like product data management (PDM) help manage design data and version control across multiple iterations of a product. Along with a scientific background, scientific communication for a wide range of audiences is another important skill which industries are looking for in order to spread awareness about the technology. It was found that about 30.8% (12 mentions) was about excellent written and verbal communication, along with teamwork/interpersonal communication.

4.3.2 Desired qualifications in spin qubits

The study of semiconductors forms a part of solid state physics and, since it is used heavily in modern electronics, it is also part of different engineering curricula across the world [71]. The qualifications (Total = 18 mentions) required for this industry span both physics and engineering. From Fig. 6, around 55.6% (10 mentions) of job descriptions ask for an electrical engineering/engineering or applied physics or computer science background. Some companies ask for more scientific backgrounds like physics, quantum optics and quantum computing, which constitutes 44.4% (8 mentions).

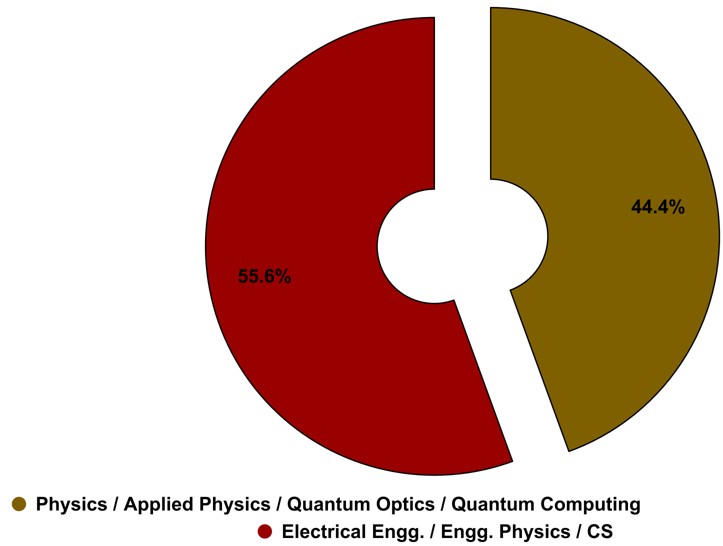

Figure 6 Desired qualifications in spin qubits

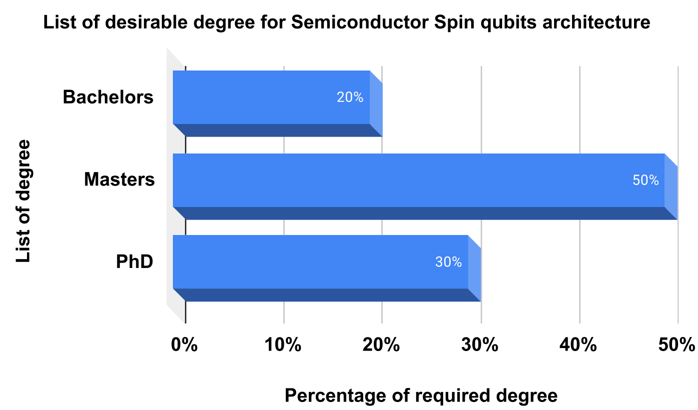

Figure 7 Degrees required for spin qubits

4.3.3 List of desired degree in semiconductor spin qubit architecture

From Fig. 7, most of the job descriptions (total mentions = 20) require the candidates to have a master's degree at least in one of the qualified areas discussed in the previous section, where it was mentioned 10 times (50%). For certain technical jobs that require higher expertise, such as integrated photonics or handling complex laboratory equipment such as vacuum or cryogenics, a PhD degree was said to be required (6 mentions or 30%).

For the jobs with only a bachelor's degree (4 mentions or 20%), typically they need some relevant working experience of about 3 to 7 years (especially clean room knowledge) for the skills which it is hiring for.

4.4 Nitrogen vacancy centre in diamond

Currently, there is a very small number of companies that are exploring this technology, which makes the jobs limited, and some of them do not have any new vacant positions. Most of the research in this field is restricted to academic institutions. However,

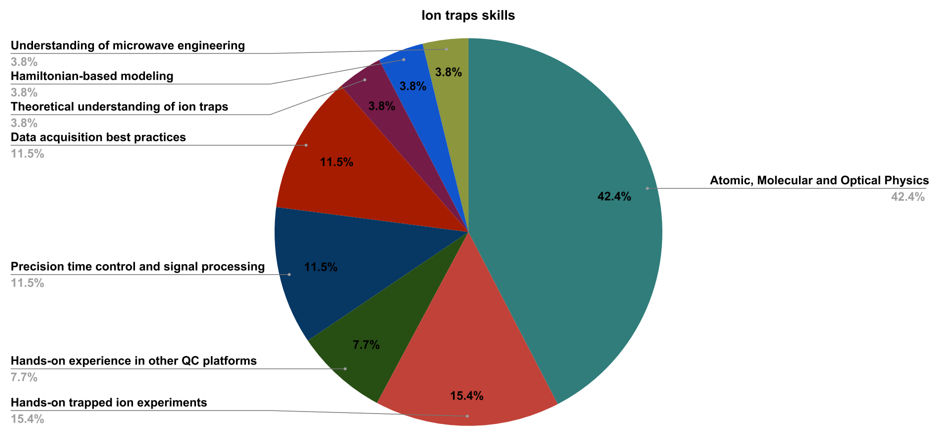

Figure 8 Trapped Ion skills for Industry roles

by analysing the available roles, we found that the industrial skills required to work in nitrogen vacancy centres are similar to those of semiconductor spin qubits, with additional skills required in confocal microscopy and laser spectroscopy. Depending on the type of job, some fabrication experience related to chemical vapour deposition and material characterisation techniques such as AFM (Atomic Force Microscopy), XRD (X-ray Diffraction), and XPS (X-ray Photoelectron Spectroscopy) are helpful.

4.5 Trapped ion qubits

4.5.1 List of desired skills in trapped ion

From Fig. 8, we can understand that the expertise in ion traps involves a deep understanding of a varied field. There were 13 job descriptions and 26 mentions of different skills. Starting with the foundational knowledge in atomic, molecular and optical Physics (11 mentions or 42.4%) is mandatory because it provides theoretical understanding of ion traps (1 mention or 3.8%) and experimental understanding of ion behaviours, interactions, laser-atom interactions, and quantum state manipulation, forming the core principle behind trapped-ion quantum systems. This theoretical foundation gives hands-on knowledge in trapped-ion experiments (4 mentions or 15.4%). This experience is useful for performing experimental realisation on other hardware platforms (2 mentions or 7.7%), such as neutral atoms.

In addition, skills in precision time control and signal processing (3 mentions or 11. 5%) are crucial, as precise control over lasers and RF fields determines the reliability and accuracy of quantum gates and measurements. Currently, best practices for data acquisition (3 mentions or 11.5%) refer to the standardised methods and strategies used to reliably collect, manage and analyse experimental data in a scientific setting. Some of the commonly used tools and systems for data acquisition practices include FPGA-based control systems, time-to-digital Converters (TDC), photo-detectors, Lab View, Python, ARTIQ (advanced real-time infrastructure for quantum physics), HDF5 (For storing large experimental datasets), etc. Also, understanding of microwave engineering (1 mention or 3.8%) is important because in trapped ion systems, microwave engineering enables precise, stable, and scalable control of qubit states using hyperfine or Zeeman transitions.

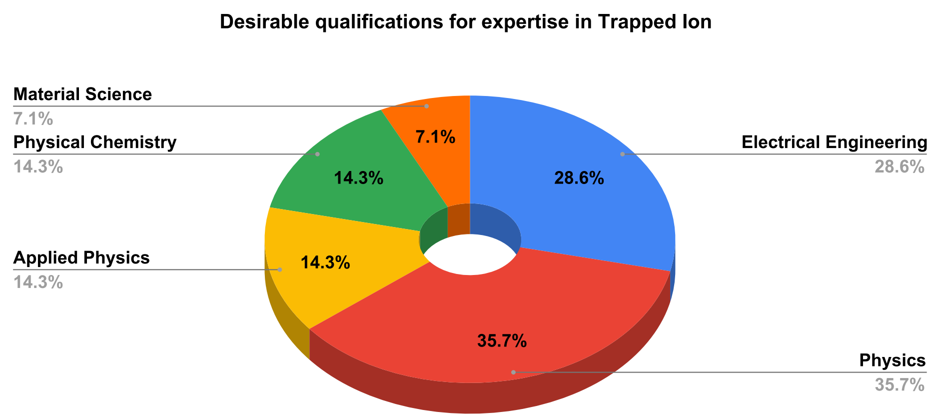

Figure 9 List of qualifications for expertise in Trapped Ion

4.5.2 List of qualifications for trapped ion qubits expertise

From Fig. 9 (total mentions = 14), we can understand that physics, physical chemistry, applied physics, material science (1 mention or 7.1%) and electrical engineering are the most sought-after qualifications expected by Industry roles. Physics (5 mentions or 35.7%) and applied physics (2 mentions or 14.3%) provide the fundamental understanding of quantum mechanics, electrodynamics, and AMO (Atomic, Molecular and Optical) physics, which are crucial for ion trapping, laser cooling, and qubit manipulation. Physical chemistry (2 mentions or 14.3%) is the study of matter's behaviour at the atomic, molecular, and subatomic levels using concepts from chemistry, physics, and mathematics. Physical chemists research thermodynamics, kinetics, spectroscopy, and quantum mechanics to create new theories and models that describe chemical phenomena. This knowledge is required for precise ion control, spectroscopy and interaction in trapped ion systems. Electrical engineering (4 mentions or 28.6%) plays a key role in designing and optimising RF and microwave control systems, signal processing, and FPGA-based real-time processing.

4.5.3 List of degrees in trapped ion expertise for industry roles

From Fig. 10 (26 mentions) some job descriptions specify a PhD without a specific field (7 mention or 26.9%) (i.e., indicating that a PhD in any relevant area is acceptable) whereas other job descriptions explicitly mention specific field in PhD like experimental physics (4 mentions or 15.4%), quantum computing (1 mention or 3.8%), applied mathematics (1 mention or 3.8%), or electrical engineering (1 mention or 3.8%). Also, one job description showed importance in post-doctoral experience (1 mention or 3.8%). Just like a PhD in experimental physics, the Master degree (4 mentions or 15.4%) was also preferred but with the necessary clean room and fabrication experience whereas the bachelors degree is preferred with at least 7-10 years of experience and the roles preferring the Bachelors degree (2 mentions or 7.7%) were hiring candidates in the lab technician role. Relevant experience (5 mentions or 19.2%) signifies people having a master's degree but working closely in the trapped-ion domain with at least 5 years of experience.

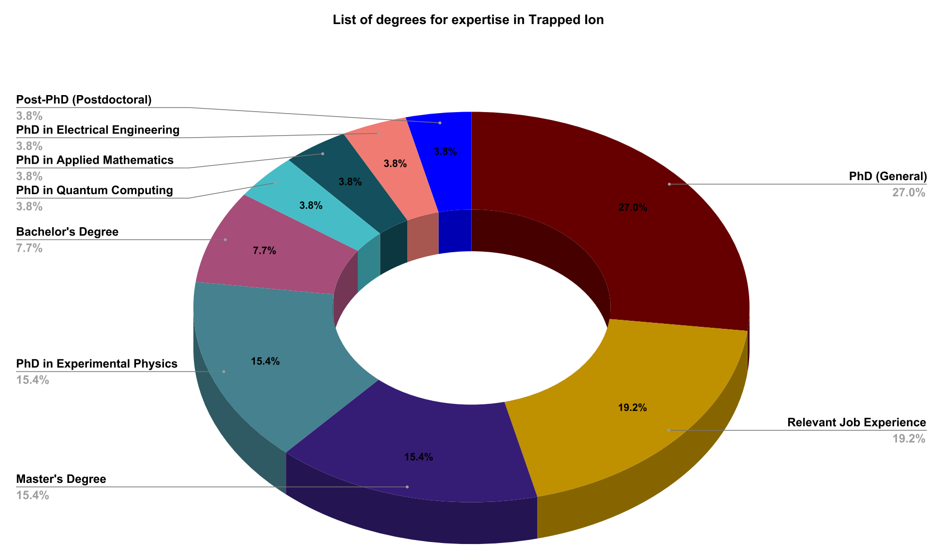

Figure 10 List of degrees for expertise in Trapped Ion

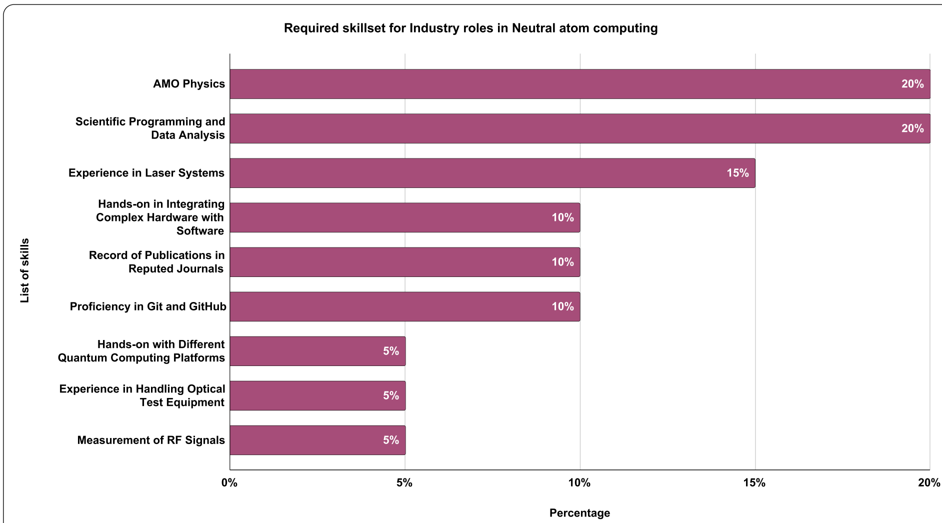

Figure 11 List of skills for expertise in Neutral atom

4.6 Neutral atom qubits

4.6.1 List of desired skills in neutral atom computing

Companies working with neutral atoms hire not only experts in neutral atoms but also laser engineers. This section has relevant fields like Laser Engineer and Optical Engineer job descriptions because of their indispensable contribution to neutral atom computing. From Fig. 11 with 20 mentions in total, there are several independent and interdependent skills required to develop expertise in neutral atom computing. A total of 3 companies and 5 job descriptions were analysed. AMO physics (4 mentions or 20%) is the foundation for understanding laser systems and optical test equipment fields. In short, atomic physics studies the interaction of atoms with other atoms and atoms with electromagnetic radiation. Molecular physics covers the study of the physical properties of molecules (groups of two or more atoms), spectroscopy, and their interactions [72]. Optical physics is the study of how light interacts with matter, with a focus on fundamental principles and new physical phenomena rather than practical applications. Investigate how electromagnetic radiation (light) interacts with atoms, molecules, and materials, resulting in phenomena such as absorption, scattering, reflection, refraction, and non-linear optical processes.

Experience with laser systems (3 mentions or 15%), such as diode lasers, optical cavities, and modulation techniques, is essential for applications in quantum computers, optical tweezers, and atomic physics research. To ensure the precision and stability of these laser-based tests, experience with optical test equipment is required. Researchers can use tools like power meters, polarimeters, beam profilers, and spectrometers to improve laser qualities, assess beam parameters, and maintain frequency stability, which is critical for cold atom investigations and trapped ion quantum computing. Scientific programming and data analysis (4 mentions or 20%) are key components of this integration, where tools such as Python, MATLAB, and LabVIEW are utilised for data collection, experiment automation, and numerical simulations of quantum systems. Understanding Git and GitHub (2 mentions or 10%) is crucial for collaborative research because of the complexity of contemporary quantum experiments. This allows for effective version control of software scripts and data analysis workflows.

Similarly, the design, implementation, and measurement of RF signals (1 mention or 5%) play a role in controlling acoustic-optic modulators or implementing spin control using RF-driven hyperfine transitions. Lastly, hands-on experience with different quantum computing platforms (1 mention or 5%) ensures adaptability and cross-platform insight, which allows researchers to draw parallels and adopt best practices from other architectures like superconducting qubits or ion traps. Ultimately, these abilities are demonstrated by a record of publications in reputable journals (2 mentions or 10%), which shows the capacity to apply the concepts of AMO physics, create experimental methods, evaluate results, and progress in quantum technology.

These abilities constitute a tightly interwoven framework, with underlying physics and coding supporting experimental control and hardware-software integration, and optics and RF ensuring physical system functionality.

4.6.2 List of desired qualifications in neutral atom computing

From Fig. 12, neutral atom computing displays 4 fields (7 mentions) and their necessary expertise. Physics (4 mentions or 57.1%) is the most out qualification because neutral-atom is built upon atomic, molecular and optical physics (AMO). Also, laser-cooling and trapping techniques such as magneto-Optical traps (MOT) and optical tweezers are essential for initialising and controlling neutral atoms. While MOT and optical tweezers are a part of atomic physics and quantum optics, their practical implementation depends on advanced optical engineering (1 mention or 14.3%). Optical engineers build and arrange lenses, mirrors, waveplates, and acousto-optic devices to manage laser intensity, focus, and polarisation, ensuring that neutral atoms are efficiently trapped and manipulated.

Neutral atom computing would be impossible without precision laser control and optical system optimisation because reliable qubit operations require highly regulated optical environments. While atomic physics and optical engineering deal with fundamental quantum interactions and laser-based manipulation, qualification in electrical engineering (1

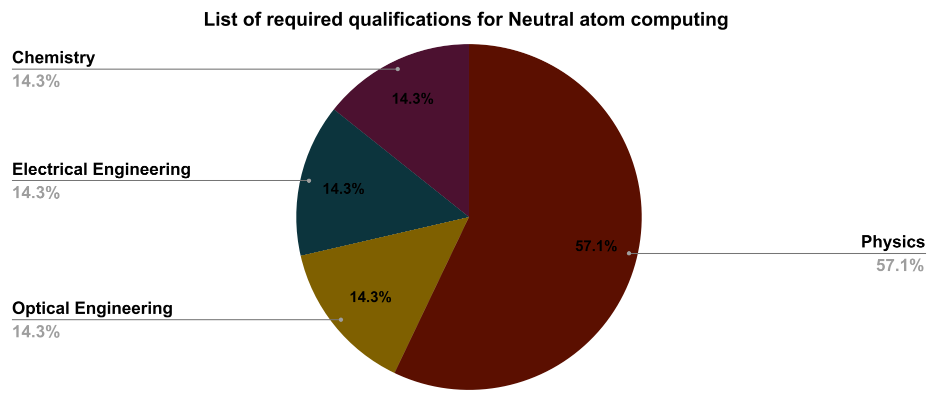

Figure 12 List of qualifications for expertise in Neutral atom

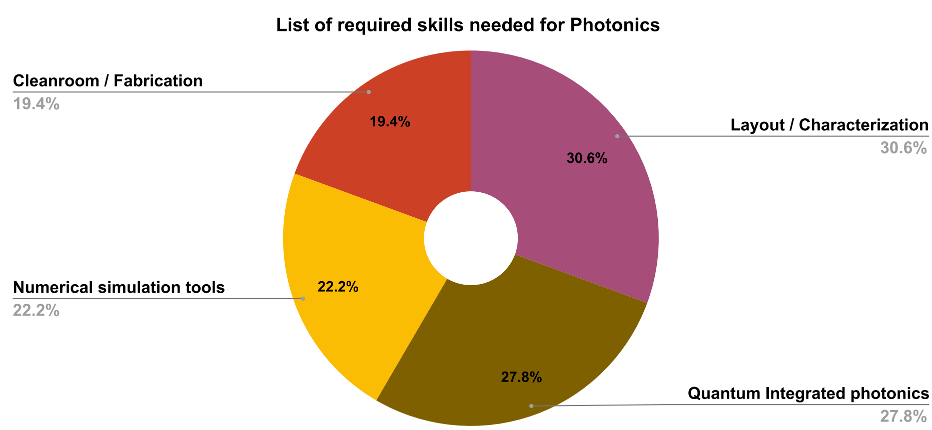

Figure 13 List of required skills for expertise in Photonics

mention or 14.3%) creates the infrastructure needed for reliable and automated functioning, such as control electronics, microwave and RF control, and detection systems. Qualifications in chemistry (1 mention or 4.3%) are desirable because of their relevance in certain areas like cold atom chemistry, material science etc.

4.6.3 List of desired degrees in neutral atom computing
Out of all the job descriptions, PhD (100%) is the only sought-after degree because neutral atom is a niche field, and deep-tech expertise is significant.

4.7 Photonics
4.7.1 List of skills required for the expertise in photonics
We went through 30 job descriptions with a total of 36 mentions of four important skills. From Fig. 13, learning numerical simulation tools (8 mentions or 22.2%) is essential for beginners in quantum photonics because they provide a practical method to understand and predict complex quantum-optical events. Simulations fill the gap between theoretical models and experimental realisations by allowing for detailed investigation of light-matter interactions, quantum states, and device performance in situations where analytical so-

lutions are generally unavailable. Numerical simulations allow beginners to investigate complex systems such as waveguides, photonic crystals, and quantum entanglement protocols without the need for expensive real prototypes. These tools also help to improve problem-solving abilities by providing insights into optimisation approaches and system-level design, both of which are critical to the advancement of quantum photonic research. Mastery of simulation tools such as Meep, COMSOL Multiphysics, and QuTiP enables learners to effectively contribute to the developing field of quantum technologies while setting the groundwork for future experimental endeavours.

Quantum integrated photonics (10 mentions or 27.8%) is the process of designing and building small photonic circuits capable of controlling single photons (light particles) in quantum states. Unlike classical photonics, which primarily deals with large numbers of photons for conventional communication or computing, Quantum integrated photonics manipulates individual photons or quantum states of photons to take advantage of quantum phenomena such as entanglement, superposition, and quantum interference. This allows for uniquely quantum-enabled tasks such as secure quantum communication, quantum computation, and precision quantum sensing beyond classical limits.

Quantum integrated photonics allows for more stable, compact, and scalable quantum systems by integrating numerous optical components, such as waveguides, beam splitters, interferometers, modulators, and single-photon detectors, onto a single silicon or semiconductor chip. Understanding the fundamentals of quantum optics, integrated photonic chip design, waveguide engineering, and quantum circuit simulation is required to learn quantum integrated photonics. This talent also works well with numerical simulation skills, allowing practitioners to effectively simulate and optimise quantum circuits prior to construction. Researchers who specialise in quantum integrated photonics are at the vanguard of new quantum technologies, making substantial contributions to fields such as quantum cryptography, quantum computer hardware, and next-generation quantum internet networks.

Clean room and fabrication experience (7 mentions or 19.4%) is critical for practically realising quantum photonic devices. A clean room is a controlled environment with extremely low levels of contaminants like dust particles, ensuring precision during fabrication processes. Quantum photonic structures frequently demand nanometer-scale accuracy, and impurities or contaminants can have a significant impact on their performance. Understanding the various lithography procedures is required when learning clean room fabrication. Clean room and fabrication experts can directly translate simulation and design ideas into real, testable quantum photonic components, bridging the gap between theoretical concepts and practical quantum technology implementations.

Designing, testing, and validating quantum optical systems requires a skill set that includes layout and characterisation (11 mentions or 30.6%) in quantum photonics. Layout is the precise design and positioning of photonic components (such as waveguides, cavities, and quantum emitters) on semiconductor wafers or integrated photonic circuits. Researchers utilise specialist software (such as KLayout, Cadence, or L-Edit) to precisely outline how components will be arranged and connected. This expertise necessitates rigorous attention to dimensions, tolerances, material qualities, and optical alignment, as even minor layout errors can dramatically impact performance. Following device fabrication, characterisation is the critical phase of experimentally evaluating and quantifying the optical and quantum properties of the photonic structures. Techniques include

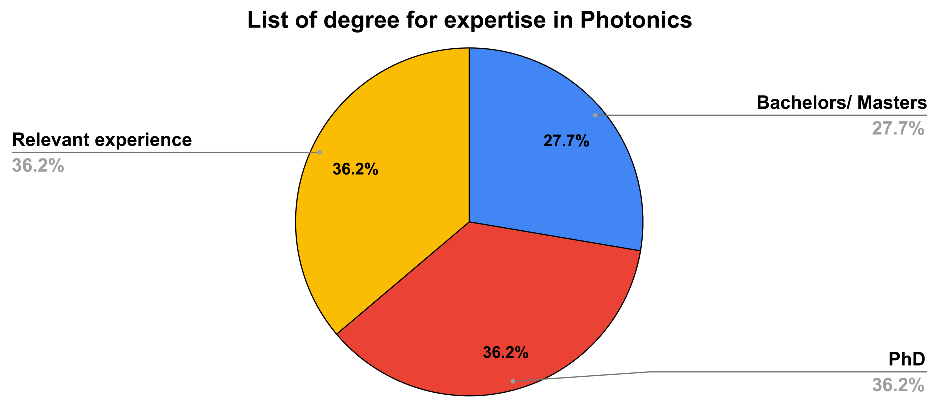

Figure 14 List of degrees for expertise in Photonics

spectroscopy (measuring photon emission and absorption), photon correlation measurements (to check single-photon behaviour), interferometry (testing quantum interference effects), and loss measurements (determining how much optical communication is maintained or lost). Layout and characterisation skills are critical for ensuring devices satisfy their intended quantum functionality, as well as troubleshooting and iteratively improving designs.

4.7.2 List of desired degree in photonics

From Fig. 14 (total mentions = 47) we can understand that a PhD (17 mentions or 36.2%) is often preferred because it signifies high level of specialized knowledge and research experience in different domains like quantum optics, photonics, electromagnetic theory, semiconductor physics, integrated photonics and nano-fabrication, signal processing and computational modelling. While bachelor's and master's degree (13 mentions or 27.7%) provides a strong educational foundation, they usually offer less exposure to the rigorous research-driven environment. And relevant research experience (17 mentions or 36.2%) in this domain means having a PhD/Master's with at least 5+ years of experience.

4.7.3 List of qualifications required for the expertise in photonics

Since this domain deals with light, the qualifications in Fig. 15 addressess different but complementary aspects of how light is used and manipulated in modern technologies (Total mentions = 26). Electrical, electronics or optical Engineering (12 mentions or 46.2%) gives a background that teaches you how to create and manage electronic circuits and optical systems, which are building blocks for devices like lasers, detectors and optical fibres. A qualification in this field can teach you how to integrate physical components that control light into the larger system that transmits it. This engineering expertise is essential when setting up the infrastructure that handles and directs light in a variety of applications. However, AMO/Applied/Quantum Physics (14 mentions or 53.9%) deals with the fundamental principles of light behaviour at atomic scales. AMO physics, which explains the phenomena that are not visible in everyday life, like how photons can exist in multiple states simultaneously or become entangled with one another. This knowledge is crucial because it builds the foundation to scale up as a photonic quantum computer.

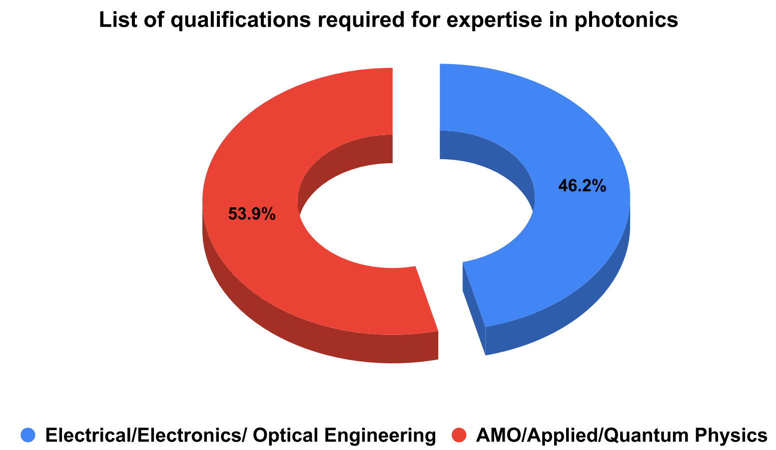

Figure 15 List of qualifications for expertise in Photonics

4.8 Quantum machine learning and AI

4.8.1 Desired degree and qualifications in quantum machine learning

Quantum machine learning as a field is in the nascent stages of development, and due to this, there are limited roles available. These are mostly application-based in the sense that the variational quantum circuits are applied for a particular application, like chemistry or finance. Careers in the field of quantum machine learning usually require a master's degree or a PhD in the domain of either computer science, chemistry, physics, engineering or applied mathematics [73].

4.8.2 Skills required for expertise in quantum machine learning

Since QML is closely inspired by classical machine learning, therefore, knowledge of applied AI algorithms, large-scale machine learning deployment, deep learning and execution of them in high-performance computing clusters (multi-GPU, distributed multi-node) is preferred. The field of QML is in the early research phase and not as mature as classical ML. As a consequence, industries also require publications (either conference or journal) which show the independent research capabilities of a candidate. Similar to other fields, knowledge of quantum computing, quantum physics or quantum many-body physics is helpful; however, different industries have different requirements based on the applications they focus on.

4.9 Quantum finance

There was only one company (with 1 job description) which was actively hiring in the quantum finance domain. Since not many details were available, we present you with some important skills required. Candidates must be able to effectively connect the creation of quantum algorithms, especially in optimisation and quantum machine learning, with real-world applications on quantum hardware in order to succeed in quantum finance. Peer-reviewed papers showcasing research contributions demonstrate the theoretical rigour and inventive potential needed in this new discipline. Typically, the Python programming language is used for financial data analysis and algorithmic development, but recently Julia and MATLAB have also been used and are helpful in carrying out scientific computing for quantum finance. Another important area for finance companies is security because for which there is also a focus on post-quantum cryptography and quantum key distribution algorithms. Some other skills that are helpful in this domain are data science, high-performance computing and standard finance knowledge, which put professionals in a unique position to promote the strategic and useful advantages of quantum computing.

4.10 Quantum cryptography

We went through several startups, and only a few of them had a dedicated career page, and within that few, only 5 companies had a designated career page with relevant job postings in the quantum cryptography domain. Some companies had one or more open vacancies, but few companies had only one vacancy to fill. In total, we analysed 14 job descriptions. After careful observation, it's clear that quantum cryptography has hardware as well as software opportunities. We divided the skill set expectations into two categories: hardware and software skills. Each category holds a variety of skills with varied mentions.

4.10.1 Desired hardware expertise in quantum cryptography by industry

The analysis of 14 job descriptions for hardware roles in quantum cryptography yielded a total of 31 mentions of various skill sets. Fig. 16 explains that the most emphasised expertise is FPGA tools and workflows(8 mentions or 25.8%), which are critical for creating high-performance cryptographic protocols and expediting hardware development. Hardware description languages such as Verilog and VHDL (4 mentions or 12.9%) are closely linked with FPGA use, allowing for exact hardware design. RF design (3 mentions or 9.7%) and quantum random number generators (QRNG's) (3 mentions or 9.7%) are essential for signal transmission and generation of secure random keys, particularly in quantum key distribution (QKD) systems. ASIC development (2 mentions or 6.5%) is required for producing compact, efficient, and scalable cryptographic hardware, whereas RSA, ECC, and AEC (2 mentions or 6.5%) demonstrate the integration of conventional cryptography into quantum systems. Similarly, post-quantum cryptography (PQC) (2 mentions or 6.5%) focuses on the transition to quantum-resistant methods. Expertise in hardware security solutions (e.g., HSM's, TPM's, 2 mentions or 6.5%) ensures secure key storage and processing, while low-noise analogue/mixed signal design (2 mentions or 6.5%) enables precise

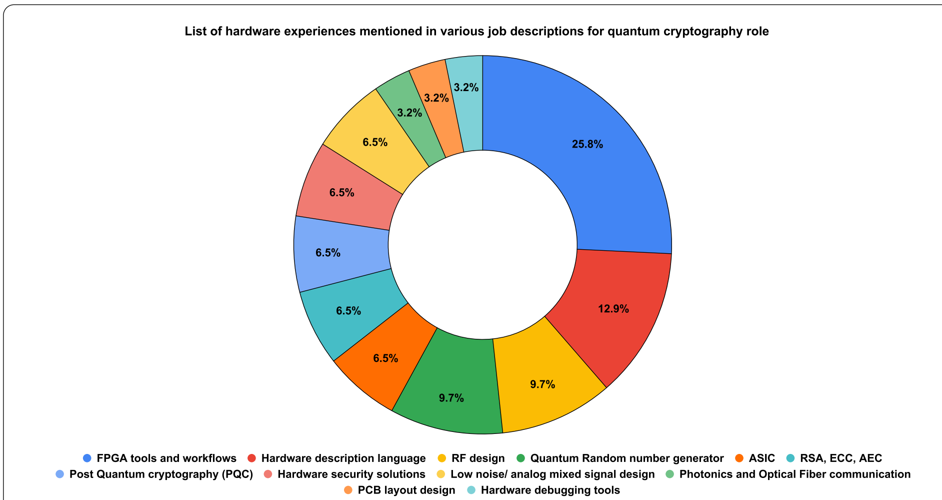

Figure 16 Skill set in hardware implementation of quantum cryptography

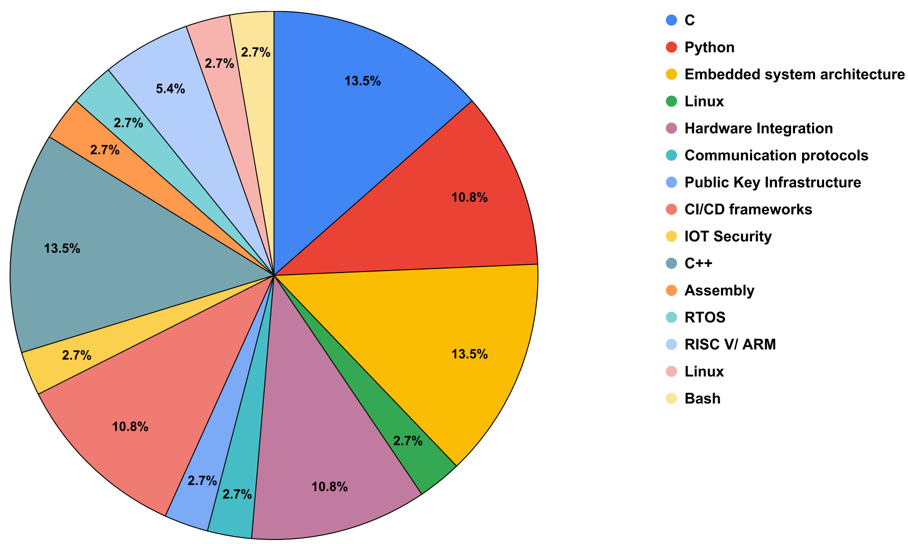

Figure 17 List of software skills for quantum cryptographic role

signal processing in quantum devices. Photonics and optical fibre communication (1 mention or 3.2%) are required for quantum signal transmission, whilst PCB layout design (1 mention or 3.2%) and hardware debugging tools (1 mention or 3.2%) help with hardware dependability and debugging. These interconnected talents demonstrate the broad competence needed to create secure, efficient, and scalable quantum cryptography systems.

4.10.2 Desired software expertise in quantum cryptography by industry

From Fig. 17, the mentioned skills (37 mentions) are foundational and advanced software technologies and practices that are important in cryptography and cybersecurity. Knowledge in programming languages like C (5 mentions or 13.5%), C++ (5 mentions or 13.5%), and Python (4 mentions or 10.8%) provide essential coding abilities for developing cryptographic algorithms, tools, and systems. Having sound Knowledge in Embedded Systems (5 mentions or 13.5%) and hardware integration (4 mentions or 10.8%) is crucial because cryptographic algorithms often run on specialised hardware to enhance security and performance. Also, having familiarity with operating systems, specifically Linux (1 mention or 2.70%), is necessary as it's widely used for security applications due to its openness and customisation capabilities. Skills in communication protocols (TLS, VPN's) (1 mention or 2.7%) and public key infrastructure (PKI) (1 mention or 2.7%) are critical, as these enable secure data transmission and authentication across networks. Practices such as CI/CD frameworks (4 mentions or 10.8%) ensure secure, automated software deployments, reducing vulnerabilities. Additionally, familiarity with IoT security (1 mention or 2.7%) and the low-level concepts of assembly (1 mention or 2.70%), RTOS (1 mention or 2.7%), and RISC-V/ARM (1 mention or 2.7%) architectures is crucial for securing resource-constrained devices. Finally, hands-on in Linux command-line (Bash) (1 mention or 2.7%) helps manage secure configurations and perform security assessments efficiently.

In short, these skills are interdependent in nature: programming languages and embedded knowledge allow cryptographic implementation. Linux and Bash skills facilitate sys-

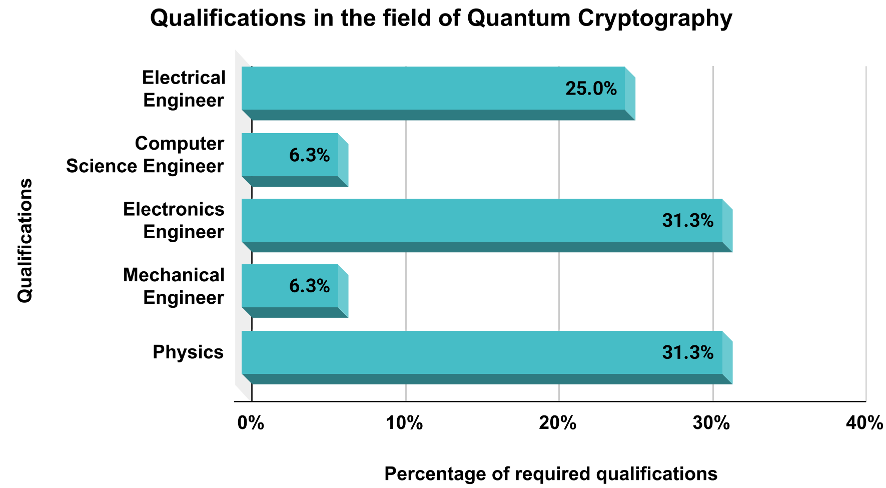

Figure 18 List of desired qualifications in the quantum cryptography field

tem management. PKI and communication protocols enable secure communication, while knowledge of hardware architectures and assembly ensures efficient and secure hardware-software integration, thereby providing a comprehensive skill set essential for effective cryptography practice.

4.10.3 List of desired qualifications in quantum cryptography field

By careful analysis, there were 16 mentions of qualifications desired by the industry. Fig. 18 explains the importance of electrical (5 mentions or 31.3%), computer science (5 mentions or 31.3%) and electronics engineer (4 mentions or 25.0%) to pursue a career in the quantum cryptography field. This is because electrical engineers are responsible for designing critical electrical systems, such as signal processing, power management, and noise reduction, to ensure that quantum hardware, such as QKD devices, operates efficiently and at high speeds. Computer scientists create cryptographic algorithms, safe communication protocols, and software stacks for controlling quantum systems that smoothly integrate classical and quantum components.

Electronics engineers specialise in hardware prototyping, embedded system development, and improving FPGA and ASIC designs to speed up cryptographic computations and ensure scalability. Mechanical engineers (1 mentions or 6.3%) play an important role in guaranteeing the mechanical integrity of quantum devices by designing tough enclosures and addressing issues like thermal dissipation and vibration in hostile environments. Finally, physicists (1 mentions or 6.3%) lay the theoretical groundwork, expanding on quantum mechanics concepts to create protocols like BB84 and technology like QRNG's. These fields work together to handle the trans-disciplinary challenges of quantum cryptography by linking theory, hardware, and software to construct robust and scalable systems.

4.10.4 List of desirable degrees for industry for quantum cryptographic role

There were a total of 7 mentions of different degrees in the job description. Fig. 19 explains the industry standards of hiring master's graduates (42.9% or 3 mentions) over bachelor's (28.6% or 2 mentions) and PhD (28.6% or 2 mentions) with required expertise. But

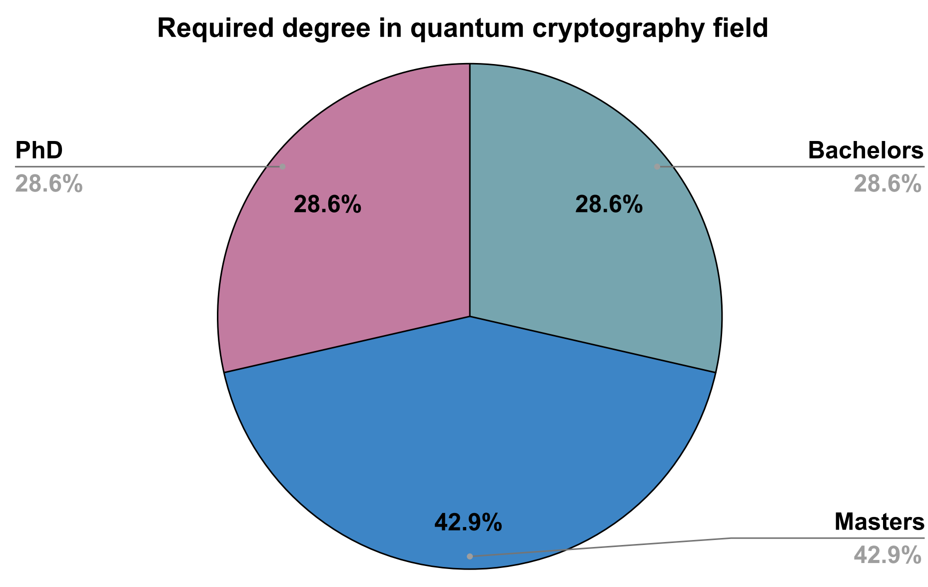

Figure 19 List of desired degrees required by the industry in the quantum cryptography field

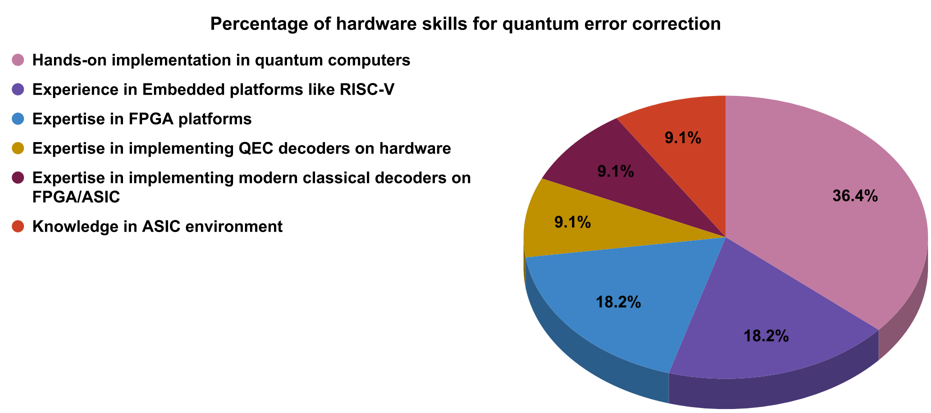

Figure 20 Percentage of hardware skills for Quantum error correction

some job requirements didn't mention any degree specification. It's clear that if a person is aligned to do pure research, then a PhD can be a solid qualification. Otherwise, a master's degree or a bachelor's degree (in some role) is satisfactory.

4.11 Quantum Error Correction (QEC)

We analysed 5 companies and 10 job descriptions in total. Companies were showing multiple job offerings. After careful analysis, we segregated the skills into hardware-based and software-based, with varied mentions for each category.

4.11.1 Desired hardware expertise in QEC by industry

There were a total of 11 mentions of hardware skills. From Fig. 20 we can understand that the hardware skills necessary for QEC are intricately linked, creating a pipeline from theory to practical application. Hands-on implementation in quantum computers (4 mentions or 36.4%) is required for applying and evaluating error correction protocols in real-world situations. Embedded platforms, such as RISC-V (2 mentions or 18.2%), provide control and interface methods, whereas FPGA platforms (2 mentions or 18.2%) are used to prototype and test QEC algorithms. FPGAs are critical for developing and improving both QEC decoders (1 mention or 9.1%) and current classical decoders (1 mention or 9.1%) like LDPC and Turbo codes. These decoders manage real-time error correction in quantum calculations.

After prototyping is complete, ASIC environments (1 mention or 9.1%) are utilised to create optimised, scalable hardware for QEC tasks. Embedded systems and ASICs collaborate to guarantee optimal integration and operation. This method combines theoretical error correction with practical implementation, making these abilities essential for advancing fault-tolerant quantum computing.

4.11.2 Desired software expertise in QEC by industry

There were a total of 30 mentions of various software skills. From Fig. 21, the industry-relevant software skills in QEC are interdependent in nature, establishing a unified framework that connects theory, simulation, and practical application. This framework is built around high-performance decoders (4 mentions or 13.3%) and fault-tolerant architectures (3 mentions or 10%), which allow for real-time error detection (3 mentions or 10%) and repair while maintaining robustness during quantum processes. These use numerical simulations and error-corrected logic to validate and optimise designs prior to hardware deployment (3 mentions or 10%). Python (3 mentions or 10%) and C++ programming skills (3 mentions or 10%) are required for designing QEC algorithms, simulating protocols, and integrating high-performance software with real-time systems (2 mentions or 6.7%) that can respond to problems dynamically.

Classical and quantum decoding approaches (2 mentions or 6.7%) apply classical error correction methods to quantum systems, whereas quantum machine learning algorithms

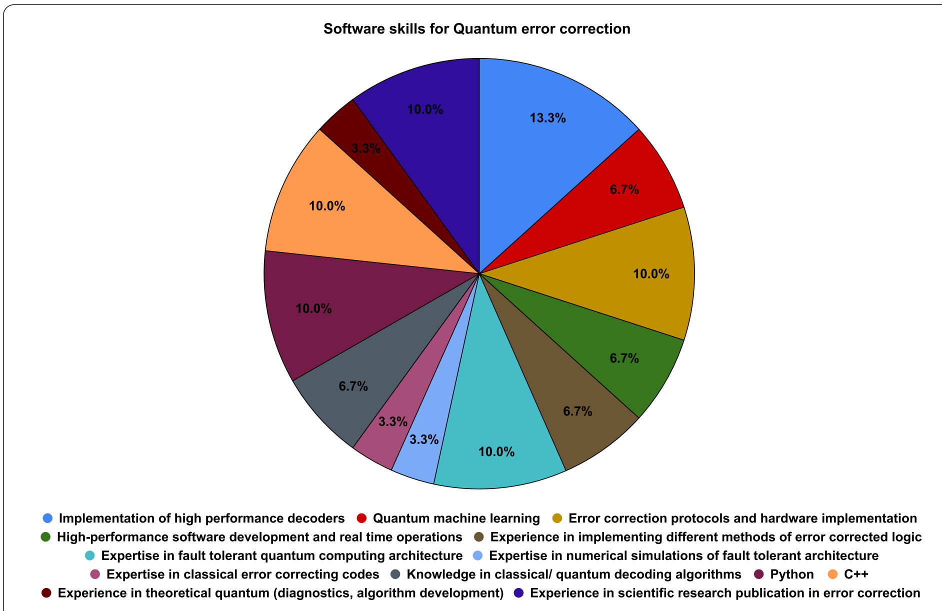

Figure 21 Percentage of software skills for Quantum error correction

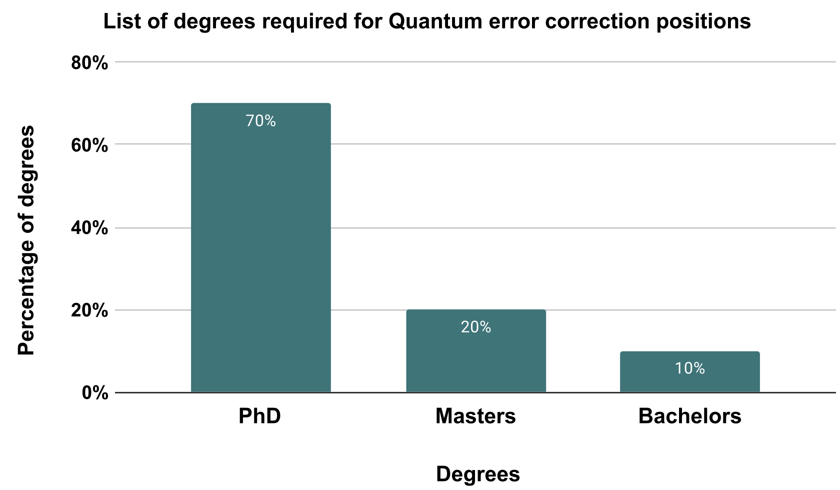

Figure 22 Degree requirements for Quantum error correction position

(2 mentions or 6.7%) enhance decoder designs and error prediction procedures, which are closely related to numerical simulations. Theoretical quantum mechanics (1 mentions or 3.3%) provides the fundamental concepts for creating decoders, fault-tolerant systems, and error-correction protocols (1 mentions or 3.3%), which are supplemented by knowledge of classical error-correcting coding (1 mentions or 3.3%). Together, these abilities provide a well-integrated ecosystem in which theoretical insights, programming experience, and advanced computational tools collaborate to promote the development and implementation of quantum error correction algorithms.

4.11.3 List of desirable degrees for industry for QEC role

There were a total of 10 mentions of the 3 major degrees. From Fig. 22, we can observe that most of the job descriptions ask for a PhD (7 mentions or 70%) due to the technicality of the error correction subject, which needs specific expertise. Master's degree (2 mentions or 20%) and only one job description showed interest in hiring a bachelor's degree holder (1 mention or 10%).

4.11.4 Industry desired qualifications for quantum error correction position

From Fig. 23 we can observe (total mentions = 37) that a qualification in computer Science (8 mentions or 21.6%) is the most sought after in industry roles, whereas physics and maths qualifications have 6 mentions or 16.2%. Some roles hire candidates who have completed a PhD or master's in quantum science and technology (5 mentions or 13.5%). Some companies specifically mentioned candidates who have done their PhD in quantum error correction (3 mentions or 8.1%) and electrical engineer qualifications had 3 mentions or 8.1%. The *equivalent professional experience* (3 mentions or 8.1%) safely assumes those candidates who have not done a PhD from any of the above-mentioned qualifications but have a master's with a track record of publications in high-impact journals and conferences, achieving an in-depth understanding of fundamental concepts and hands-on experience.

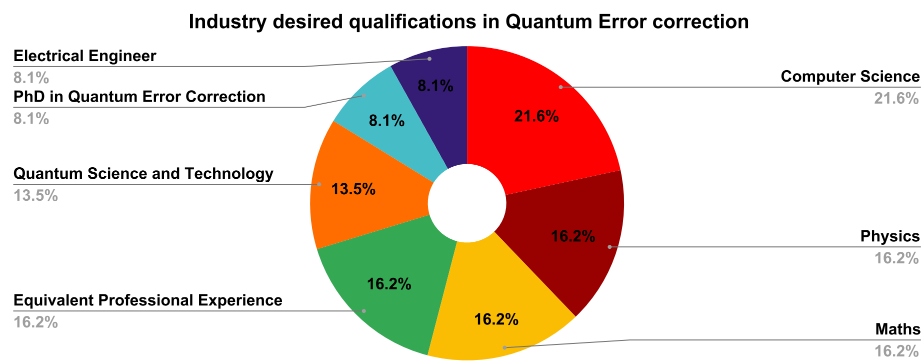

Figure 23 Industry desired qualifications for QEC position

4.12 Quantum sensing

Quantum sensing is actively pursued by various governments across different countries as it finds its applications in defence and the military. Due to this reason, job availability is scarce, and here, we discuss the industrial requirements based on three job descriptions offered by a single company.

4.12.1 Skills

The skills required in the field of quantum sensing are diverse and depend on the type of role the candidate is applying for and however, there are some general skills to have for this role. Experience in magnetic/gravitational-based surveying, motional characterisation, and signal processing is helpful. It is also required to have knowledge on cold atoms, laser system design, testing, atomic optics and optical/scientific device design and construction and microwave synthesis and control for digital hardware and software systems. Python, C++ or C language are usually preferred for systems programming and data processing, with experience in continuous integration/deployment and best software practices, adding more value.

4.12.2 Qualifications

For quantum sensing positions, typically, the candidate should have qualifications in either physics, applied physics, engineering, data science or geophysics-related disciplines along with some relevant industrial experience.

4.12.3 Degrees

A PhD is a requirement in most of the job descriptions. However, if a candidate has an appropriately qualified master's degree, then companies also value relevant industrial experience with the master's.

4.13 Quantum metrology

Quantum metrology is the most undervalued area of science. Metrology is different from meteorology, in which the latter studies atmospheric science. Metrology is the science of measurements essential for determining quantum states. It is a discipline dedicated to the accurate measurement of physical characteristics, including phase, time, magnetic fields, and other tangible quantities, utilising quantum phenomena such as entanglement and squeezing [74]. The primary objective of quantum metrology is to enhance measurement precision beyond conventional thresholds, including the standard quantum limit (SQL), by utilising the distinctive characteristics of quantum states [75]. It quantifies continuous physical characteristics, frequently incorporating variables like as phase shifts, magnetic field intensities, or time intervals, with exceptional accuracy [76]. These measurements utilise quantum states that exhibit heightened sensitivity to minor environmental alterations, rendering quantum metrology essential for applications such as atomic clocks [77], gravitational wave detection, and magnetometry. The primary objective is to improve measurement precision for practical applications, minimising measurement uncertainty by the utilisation of quantum phenomena, thus enhancing accuracy in fields such as navigation, telecommunications, environmental monitoring, and healthcare [78].

Quantum metrology measurements generally yield continuous values that denote physical parameters, including phase, frequency, or magnetic field intensity. These values undergo classical post-processing to produce highly accurate estimations. Quantum metrology finds its main applications in sensing and precision measurement, including ultra-precise atomic clocks for GPS and telecommunications, gravitational wave detectors using quantum-enhanced interferometry, quantum magnetometers for sensitive magnetic field measurements and quantum sensors for applications in healthcare [79], navigation, and environmental monitoring. This makes quantum metrology essential for advancing next-generation technologies across various sectors. One should note that there are very few companies working on quantum metrology, and the roles are limited at present. Scientists often view it as an understudied field.

5 Conclusion

In conclusion, our critical analysis of industry job descriptions across diverse qubit platforms—from superconducting and semiconducting devices to ion traps, neutral atoms, NV centres, topological approaches, and photonic systems—alongside emerging software arenas like quantum machine learning, cryptography, and error correction, reveals both common and modality-specific requirements. Based on the companies considered in this paper, employers seek strong foundations in quantum physics, applied mathematics, and algorithm design, complemented by proficiency in programming (e.g., Python, Qiskit, Cirq) and hands-on laboratory or simulation experience. At the same time, each hardware modality is going to necessitate specialised training—e.g., cryogenics for superconducting qubits or vacuum and laser control for ion traps—and software jobs demand experience with high-performance computing and formal verification. Based on the job descriptions surveyed, we observed that quantum hardware, error correction and cryptography-related roles dominate as compared to software roles such as finance and quantum machine learning. This phenomenon can be attributed to the need and priority of companies to develop a fault-tolerant quantum computer that can be utilised to solve a real-world and impactful complex problem. It is worthwhile to note that quantum machine learning roles are advertised as a skill in internal R&D of a company's project and sometimes do not appear in the mainstream career page. To close these gaps, schools must adopt flexible, modular curricula that pair core quantum science with specialised tracks (hardware, software, sensing) and develop partnerships with industry and national laboratories to provide internships, test bed access, and capstone projects. Moreover, incorporating training in control engineering, error mitigation, and cross-disciplinary collaboration will prepare graduates to work in the team-based environment of quantum

R&D. Based on the comprehensive analysis of various industrial roles, we provide suggestions for candidates who would like to pursue a career in quantum technology.

We recommend a few steps to help students or industry professionals develop an industry-desired skill set - *Make quantum interesting:* As pointed out in [11], students assume that quantum mechanics is too complex and requires a deep understanding of maths and physics, which in one way is correct, but quantum mechanics as a subject can be pursued by anyone who is curious, interested and is not restricted to only geniuses. So, universities and schools must recruit teachers and professors who understand the essence of quantum and not just teach the subject but also instil curiosity and interest in students. In short, the willingness to learn the subject depends upon how well the subject is taught in an easier way and how curious the student becomes at the end of the year. *Online courses and certifications:* There are several online courses offered by Coursera, Udemy, The Coding School, and a few certification courses from IBM Qiskit, PennyLane, etc. *Access to quantum labs:* Future advancements in quantum theory depend upon present undergraduate/postgraduate students. Universities must develop a curriculum covering everything from basic courses to advanced courses in theoretical physics and experimental quantum physics. Lab access must be mandatory for students, and balanced knowledge should be provided in theory and experimental subjects. In this way, the gap between academia and industry can be reduced significantly.

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